KAIZEN MATHS

Lesson starter

SUVAT and Equations of Motion

Vertical Motion

Name _____

S1

A particle is projected vertically upwards with initial speed 29.4 m/s. Find the time taken to reach its greatest height. Take $g = 9.8$ m/s^2.

S2

A ball is thrown vertically upwards with speed 14 m/s. Find the greatest height above the point of projection. Take $g = 9.8$ m/s^2.

S3

A particle is projected vertically upwards with speed 14 m/s. Find its height above the point of projection after 1 s. Take $g = 9.8$ m/s^2.

S4

A ball is projected vertically upwards with speed 16 m/s. Find its speed when it is 7 m above the point of projection. Take $g = 9.8$ m/s^2.

Show your working

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MEC-8084-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

Vertical Motion

Name _____

E1

A particle is projected vertically upwards with initial speed 16 m/s. Find the time taken to reach its greatest height. Take $g = 9.8$ m/s^2.

E2

A ball is thrown vertically upwards with speed 24.5 m/s. Find the greatest height above the point of projection. Take $g = 9.8$ m/s^2.

E3

A particle is projected vertically upwards with speed 29.4 m/s. Find its height above the point of projection after 2 s. Take $g = 9.8$ m/s^2.

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A particle is projected vertically upwards with initial speed 16 m/s. Find the time taken to reach its greatest height. Take $g = 9.8$ m/s^2.

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MEC-8084-L3 · E

Show your working

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A particle is projected vertically upwards with speed 29.4 m/s. Find its height above the point of projection after 2 s. Take $g = 9.8$ m/s^2.

MEC-8084-L3 · E

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A particle is projected vertically upwards with speed 29.4 m/s. Find its height above the point of projection after 2 s. Take $g = 9.8$ m/s^2.

MEC-8084-L3 · E

Show your working

Teacher answers & working

SUVAT and Equations of Motion · Vertical Motion

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S1

A particle is projected vertically upwards with initial speed 29.4 m/s. Find the time taken to reach its greatest height. Take g = 9.8 m/s².

Answer:

$t = 3$ s

Working:

Take upward as positive and list the known values.

u = 29.4 m/s, v = 0 m/s at the greatest height, a = -9.8 m/s², t = ?

Use $v = u + at$.

$0 = 29.4 - 9.8t$

$9.8t = 29.4$

$t = 29.4 \div 9.8 = 3$

The time to greatest height is 3 s.

S2

A ball is thrown vertically upwards with speed 14 m/s. Find the greatest height above the point of projection. Take g = 9.8 m/s².

Answer:

height = 10 m

Working:

Take upward as positive and list the known values.

u = 14 m/s, v = 0 m/s at the greatest height, a = -9.8 m/s², s = ?

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute and solve for s.

$0^2 = 14^2 + 2 \times (-9.8) \times s$

$0 = 196 - 19.6s$

$19.6s = 196$

$s = 196 \div 19.6 = 10$

The greatest height is 10 m.

Teacher answers & working

SUVAT and Equations of Motion · Vertical Motion

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S3

A particle is projected vertically upwards with speed 14 m/s. Find its height above the point of projection after 1 s. Take g = 9.8 m/s².

Answer:

height = 9.1 m

Working:

Take upward as positive and list the known values.

$u = 14$ m/s, $a = -9.8$ m/s², $t = 1$ s, $s = ?$

Use $s = ut + \frac{1}{2}at^2$.

$s = 14 \times 1 + \frac{1}{2} \times (-9.8) \times 1^2$

$s = 14 - 4.9 = 9.1$

The height is 9.1 m above the point of projection.

S4

A ball is projected vertically upwards with speed 16 m/s. Find its speed when it is 7 m above the point of projection. Take g = 9.8 m/s².

Answer:

speed = 10.9 m/s

Working:

Take upward as positive and list the known values.

$u = 16$ m/s, $a = -9.8$ m/s², $s = 7$ m, $v = ?$

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute the values.

$v^2 = 16^2 + 2 \times (-9.8) \times 7$

$v^2 = 256 - 137.2 = 118.8$

$|v| = \sqrt{118.8} = 10.9$

The speed is 10.9 m/s. The velocity could be upward or downward at that height.

Teacher answers & working

SUVAT and Equations of Motion · Vertical Motion

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E1

A particle is projected vertically upwards with initial speed 16 m/s. Find the time taken to reach its greatest height. Take g = 9.8 m/s².

Answer:

$t = 1.63$ s

Working:

Take upward as positive and list the known values.

u = 16 m/s, v = 0 m/s at the greatest height, a = -9.8 m/s², t = ?

Use $v = u + at$.

$0 = 16 - 9.8t$

$9.8t = 16$

$t = 16 \div 9.8 = 1.63$

The time to greatest height is 1.63 s.

E2

A ball is thrown vertically upwards with speed 24.5 m/s. Find the greatest height above the point of projection. Take g = 9.8 m/s².

Answer:

height = 30.62 m

Working:

Take upward as positive and list the known values.

u = 24.5 m/s, v = 0 m/s at the greatest height, a = -9.8 m/s², s = ?

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute and solve for s.

$0^2 = 24.5^2 + 2 \times (-9.8) \times s$

$0 = 600.25 - 19.6s$

$19.6s = 600.25$

$s = 600.25 \div 19.6 = 30.62$

The greatest height is 30.62 m.

Teacher answers & working

SUVAT and Equations of Motion · Vertical Motion

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E3

A particle is projected vertically upwards with speed 29.4 m/s. Find its height above the point of projection after 2 s. Take g = 9.8 m/s².

Answer:

height = 39.2 m

Working:

Take upward as positive and list the known values.

$u = 29.4$ m/s, $a = -9.8$ m/s^2, $t = 2$ s, $s = ?$

Use $s = ut + \frac{1}{2}at^2$.

$s = 29.4 \times 2 + \frac{1}{2} \times (-9.8) \times 2^2$

$s = 58.8 - 19.6 = 39.2$

The height is 39.2 m above the point of projection.